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Class : 12

Register Number

COMMON HALF YEARLY EXAMINATION,2025-26

Time Allowed : 3.00 Hours] **CHEMISTRY** [Max. Marks : 70

PART – I

I. Choose the correct answer. 15x1=15

1. Extraction of gold and silver involves leaching with cyanide ion silver is later recovered by
   a) Distillation b) Zone refining c) Displacement with zinc d) liquation
2. Duralumin is an alloy of
   a) Cu, Mn b) Cu, Al, Mg c) Al, Mn d) Al, Cu, Mn, Mg
3. The salt of nitric acid, which is used in photography
   a) $AgNO_3$ b) $NaNO_3$ c) $Mg(NO_3)_2$ d) $Cu(NO_3)_2$
4. **Assertion** : Ce^{4+} is used as an Oxidising agent in Volumetric Analysis.
   Reason : Ce^{4+} has the tendency of attaining +3 Oxidation states.
   a) Both Assertion and Reason are true and reason is the correct explanation of Assertion.
   b) Both Assertion and Reason are true but reason is not the correct explanation of Assertion
   c) Assertion is True but Reason is False d) Both Assertion and Reason are False.
5. In $[Fe(H_2O)_6]^{3+}$, the electronic configuration of central metal atom is
   a) $t_{2g}^3 e_g^2$ b) $eg^3 t_{2g}^3$ c) $t_{2g}^2 e_g^2$ d) $t_{2g}^4 e_g^3$
6. In calcium fluoride having the flurite structure, the Coordination number of Ca^{2+} ion and F^- ion are
   a) 4 and 2 b) 6 and 6 c) 8 and 4 d) 4 and 8
7. In a first order reaction $x \rightarrow y$; if k is the rate constant and the initial concentration of the reactant x is 0.1 M then, the half life is
   a) $\left(\frac{\log 2}{k}\right)$ b) $\left(\frac{0.693}{(0.1)k}\right)$ c) $\left(\frac{\ln 2}{k}\right)$ d) none of these
8. The expression for solubility product of $Ca_3(PO_4)_2$ is
   a) $9S^5$ b) $27S^6$ c) $108S^5$ d) $64S^5$
9. During electrolysis of molten sodium chloride, the time required to produce 0.1 mole of chlorine gas using a current of 3A is
   a) 55 minutes b) 107.2 minutes c) 220 minutes d) 330 minutes
10. The substance used to remove coloured impurities from sugar is
    a) Fuller's earth b) Silica gel c) Animal charcoal d) Ion Exchange resin
11. Which one of the following will react with phenol to give salicylaldehyde after hydrolysis
    a) Dichloro methane b) Trichloro ethane c) Trichloro methane d) CO_2
12. Which one of the following reduces Tollen's reagent
    a) formic acid b) acetic acid c) benzophenone d) none of these
13. The method by which aniline cannot be prepared is
    a) degradation of benzamide with Br_2/NaOH
    b) Potassium salt of Phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution
    c) Hydrolysis of Phenylcyanide with acidic solution
    d) Reduction of nitrobenzene by Sn/HCl
14. Insulin, a hormone chemically is
    a) Fat b) Steroid c) Protein d) Carbohydrates
15. Which of the following is an analgesic?
    a) Streptomycin b) Chloromycetin c) Aspirin d) Pencillin

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PART - II

II. Answer any six questions of the following. Question No. 24 is compulsory. 6x2=12

16. Write a short note on anamolous properties of the first element of p-block.
17. What are Transition metals? Give examples?
18. What are the Limitations of VB theory?
19. Give two examples for zero order reaction.
20. What is Electrophoresis?
21. Write a note on Knoevenagel reaction?
22. How are Vitamins Classified?
23. What are food Preservatives?
24. Barium has a body centered cubic unit cell with a length of 508 Pm along an edge. What is the density of barium in g cm^{-3}?

PART - III

III. Answer any six questions of the following. Question No. 33 is compulsory. 6x3=18

25. Write a note on gravity separation method?
26. What is inert pair effect?
27. Write the following for the complex $[Ag(NH_3)_2]\,Cl$
    a) IUPAC name b) Coordination number c) Nature of the complex
28. Write a note on Frankel Defect?
29. What are Buffer Solutions? Give its types?
30. Define Molar Conductance? Give its Unit?
31. How will you prepare phenol from Cumene?
32. How is Nylon 6,6 Prepared? Mention its uses?
33. A compound 'A' of molecular formula C_2H_3N on reduction with $Na(Hg)/C_2H_5OH$ gives 'B' of molecular formula C_2H_7N which undergoes Carbylamines test. Compound "B" on reaction with nitrous acid gives compound 'C' of molecular formula C_2H_6O by liberating nitrogen. Identity A, B, and C.

PART - IV

IV. Answer all the questions. 5x5=25

34. (a) (i) Explain the Electrometallurgy of Aluminium. **(5)**

    (OR)

    (b) i) Write a note on Fisher Tropsch synthesis. **(3)**
    ii) Give the Uses of Argon. **(3)**

35. (a) i) What is lanthanide contraction? What are the causes of Lanthanide Contraction. **(3)**
    ii) Write Bragg's Equation? **(2)**

    (OR)

    (b) Based on VB theory explain why $[Cr(NH_3)_6]^{3+}$ is Paramagnetic While $[Ni(CN)_4]^{2-}$ is diamagnetic.

36. (a) Define Half life of a reaction. Show that for a first order reaction half life is independent of initial concentration. **(5)**

    (OR)

    (b) i) Derive Henderson - Hasselbalch Equation. **(3)**
    ii) Can Fe^{3+} oxidises bromide to bromine under standard conditions. given $E^0_{Fe^{3+}/Fe^{2+}} = 0.771V$ $E^0_{Br_2/Br^-} = 1.09$ V **(2)**

37. (a) Differentiate Physisorption and Chemisorption. **(5)**

    (OR)

    (b) i) Explain the reducing Property of formic acid? **(3)**
    ii) Write a note on Hoffmann's degradation reaction? **(2)**

38. (a) Explain Lucas Test. **(5)**

    (OR)

    (b) i) Give any three difference between DNA and RNA. **(3)**
    ii) What are Antibiotics? **(2)**

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SACRED HEART MATRICULATION HIGHER SECONDARY SCHOOL

SHOLINGANALLUR, CHENNAI – 600119.

COMMON HALF-YEARLY EXAM – DECEMBER 2025

STD: **12** **CHEMISTRY ANSWERKEY**

TIME: **180 MINUTES** MAXIMUM MARKS: **70**

PART – A

I. ***Answer all the questions. Choose the correct answer from the given four alternatives and write the option code and the corresponding answer.*** **(15x1=15)**

1. c) Displacement with Zinc
2. d) Al, Cu, Mn, Mg
3. a) $AgNO_3$
4. a) Both Assertion and Reason are true and reason is the correct explanation of Assertion
5. a) $t_{2g}^3 e_g^2$
6. c) 8 and 4
7. c) $\frac{ln\ 2}{k}$
8. c) $108s^5$
9. b) 107.2 minutes
10. c) Animal charcoal
11. c) Trichloro methane
12. a) Formic acid
13. b) Potassium salt of Phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution
14. c) Protein
15. c) Aspirin

PART – B

II. ***Answer any six questions. Question No.24 is compulsory.*** **(6x2=12)**

16. Write a short note on anamolous properties of the first element of p-block.

In p-block elements the first member of each group differs from the other elements of the corresponding group. The following factors are responsible for this anomalous behaviour.

- Small size of the first member.
- High ionisation enthalpy and high electronegativity.
- Absence of d-orbitals in their valance shell.

The first member of the group-13, boron is a metalloid while others are reactive metals. Moreover, boron shows diagonal relationship with silicon of group -14. The oxides of boron and silicon are similar in their acidic nature.

17. What are Transition metals? Give examples?

The metallic elements that have incompletely filled d or f subshell in the neutral or cationic state are called transition metals.

Examples: Copper, Iron, Cobalt, Nickel.

18. What are the Limitations of VB theory?

- It does not explain the colour of the complex

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- It considers only the spin only magnetic moments and does not consider the other components of magnetic moments.
- It does not provide a quantitative explanation as to why certain complexes are inner orbital complexes and the others are outer orbital complexes for the same metal.

For example,

$[Fe(CN)_6]^{4-}$ is diamagnetic (low spin) whereas $[FeF_6]^{4-}$ is paramagnetic (high spin).

19. Give two examples for zero order reaction.

- Photochemical reaction between H_2 and I_2

$$H_2(g)+Cl_2(g) \xrightarrow{h\nu} 2HCl(g)$$

- Decomposition of N_2O on hot platinum surface

$$N_2O(g) \rightleftharpoons N_2(g) + \frac{1}{2}O_2(g)$$

- Iodination of acetone in acid medium is zero order with respect to iodine.

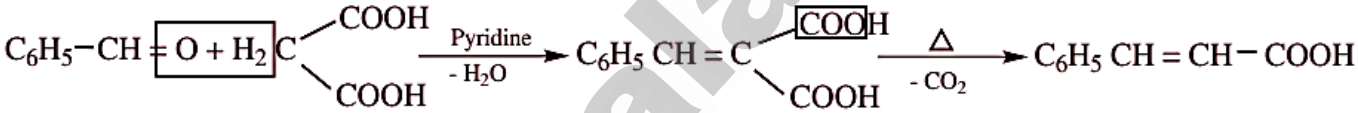
$$CH_3COCH_3 + I_2 \xrightarrow{H^+} ICH_2COCH_3 + HI$$

20. What is Electrophoresis?

This migration of sol particles under the influence of electric field is called **electrophoresis** or **cataphoresis**.

If the sol particles migrate to the cathode, then they posses positive (+) charges, and if the sol particles migrate to the anode then they have negative (-) charges.

21. Write a note on Knoevenagel reaction?



Benzaldehyde Malonic acid Cinnamic acid

Benzaldehyde condenses with malonic acid in presence of pyridine forming cinnamic acid, Pyridine act as the basic catalyst.

22. How are Vitamins Classified?

Vitamins are classified into two groups based on their solubility either in water or in fat.

Fat soluble vitamins:

These vitamins absorbed best when taken with fatty food and are stored in fatty tissues and livers. These vitamins do not dissolve in water.

Hence they are called fat soluble vitamins. Vitamin **A, D, E** & **K** are fat-soluble vitamins.

Water soluble vitamins:

Vitamins **B (B_1, B_2, B_3, B_5, B_6, B_7, B_9** and **B_{12})** and **C** are readily soluble in water. On the contrary to fat soluble vitamins, these can't be stored.

The excess vitamins present will be excreted through urine and are not stored in our body.

23. What are food Preservatives?

Preservatives are capable of inhibiting, Retarding or arresting the process of fermentation, Acidification or other decomposition of food by growth microorganisms.

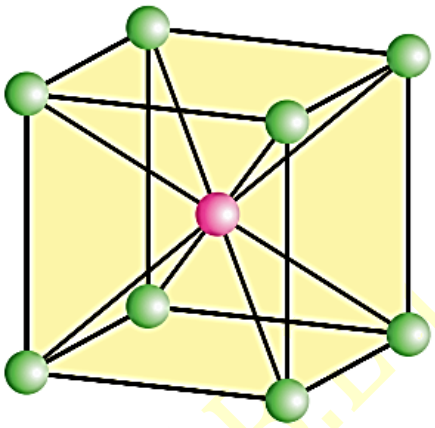
Eg.

Boric acid and Sodium benzoate.

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24. Barium has a body centered cubic unit cell with a length of 508Pm along an edge. What is the density of barium in g cm^{-3}?

$$\rho = \frac{n\,M}{a^3 N_A}$$



In this case,

n=2 ; M=137.3 $gmol^{-1}$; a = 508pm= 5.08X10^{-8}cm

$$\rho = \frac{2 \text{ atoms} \times 137.3 \text{ g mol}^{-1}}{(5.08 \times 10^{-8} \text{ cm})^3 (6.023 \times 10^{23} \text{ atoms mol}^{-1})}$$

$$\rho = \frac{2 \times 137.3}{(5.08)^3 \times 10^{-24} \times 6.023 \times 10^{23}} \text{ g cm}^{-3}$$

$$\rho = 3.5 \text{ g cm}^{-3}$$

PART – C

III. ***Answer any six questions. Question No.33 is compulsory.*** **(6x3=18)**

25. Write a note on gravity separation method?

- Gravity separation method, the ore having high specific gravity is separated from the gangue that has low specific gravity by simply washing with running water.
- Ore is crushed to a finely powdered form and treated with rapidly flowing current of water.
- During this process the lighter gangue particles are washed away by the running water.
- This method is generally applied to concentrate the native ore such as gold and oxide ores such as haematite (Fe_2O_3), tin stone (SnO_2)

26. What is inert pair effect?

In heavier post-transition metals, the **outer *s* electrons (*ns*)** have a tendency to remain inert and show **reluctance** to take part in the **bonding**. It is known as inert pair effect.

27. Write the following for the complex $[Ag(NH_3)_2]Cl$ (a) IUPAC name, (b) Coordination number & (c) Nature of the complex

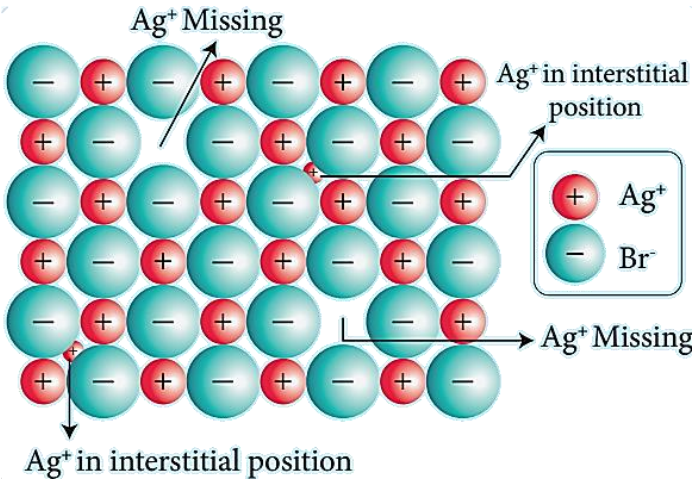
(a) IUPAC name: Diaminesilver(I)chloride

(b) Coordination number: 2

(c) Nature of the complex: Cationic Complex

28. Write a note on Frankel Defect?

- Frenkel defect arises due to the dislocation of ions from its crystal lattice.
- The ion which is missing from the lattice point occupies an interstitial position.
- This defect is shown by ionic solids in which cation and anion differ in size. Unlike Schottky defect, this defect does not affect the density of the crystal.
- For example AgBr, in this case, small Ag ion leaves its normal site and occupies an interstitial position as shown in the figure.



29. What are Buffer Solutions? Give its types?

Buffer is a solution which consists of a mixture of a weak acid and its conjugate base (or) a weak base and its conjugate acid. This buffer solution resists drastic changes in its pH upon addition of a small quantities of acids (or) bases, and this ability is called buffer action.

Basic buffer solution : a solution containing a weak base and its salt.

Example : Solution containing NH_4OH and NH_4Cl

Acidic buffer solution : a solution containing a weak acid and its salt.

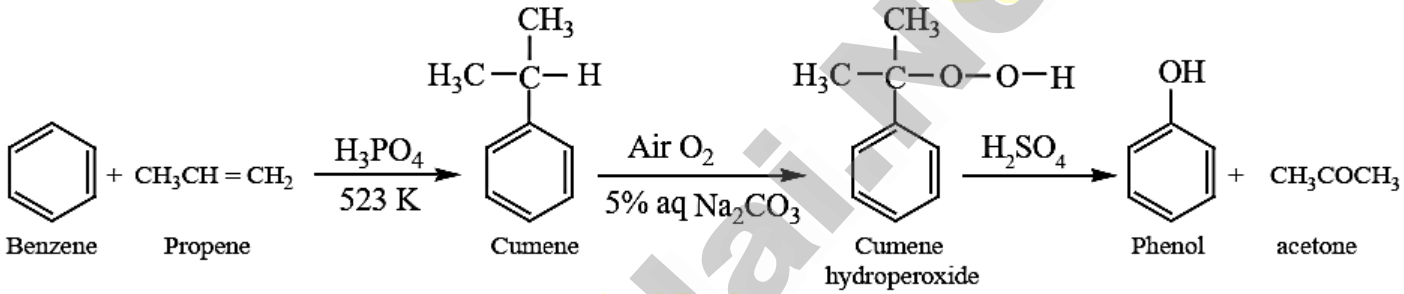
Example : Solution containing CH_3COOH and CH_3COONa

30. Define Molar Conductance? Give its Unit?

Molar conductance: The conductivity cell in which the electrodes are separated by 1m and having V m^3 of electrolytic solution which contains 1 mole of electrolyte. The conductance of such a system is called the molar conductance (Λ_m).

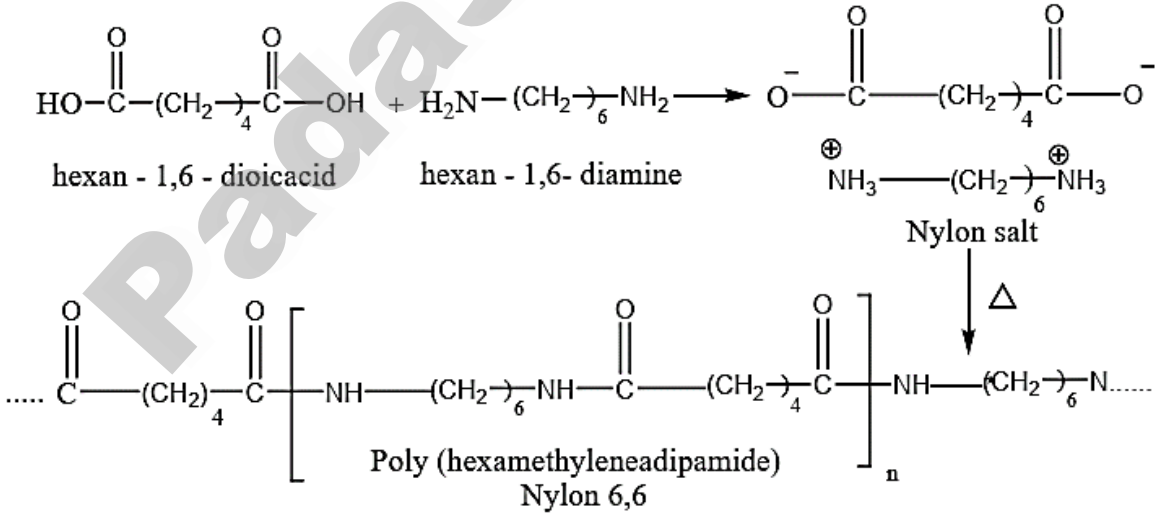
$$\Lambda_m = \frac{K\,(Sm^{-1}) \times 10^{-3}}{M}\; mol^{-1}\, m^3$$

31. How will you prepare phenol from Cumene?



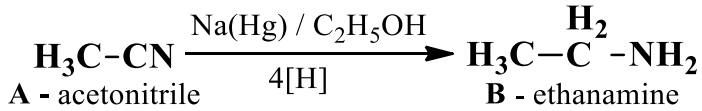
32. How is Nylon 6,6 Prepared? Mention its uses?

Nylon – 6,6 can be prepared by mixing equimolar adipic acid and hexamethylene – diamine to form a nylon salt which on heating eliminate a water molecule to form amide bonds.

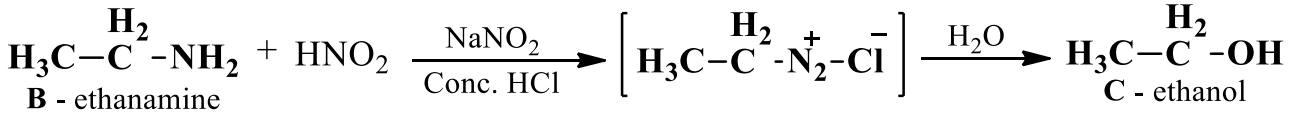


Uses: It is used in textiles, manufacture of cards etc…

33. A compound 'A' of molecular formula C_2H_3N on reaction with Na(Hg) / C_2H_5OH gives 'B' of molecular formula C_2H_7N which undergoes Carbylamines test. Compound 'B' on reaction with nitrous acid gives compound 'C' of molecular formula C_2H_6O by liberating nitrogen. Identify A, B and C.



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A – Acetonitrile (or) Methyl Cyanide;
B – Ethanamine;
C – Ethanol

PART – D

IV. ***Answer all the questions.*** **(5x5=25)**

34.(a) Explain the Electrometallurgy of Aluminium? [5Marks]

Cathode: Iron tank lined with carbon **Anode** : Carbon blocks

Electrolytes: 20% solution of alumina, obtained from the bauxite ore is mixed with molten cryolite and is taken in the electrolysis chamber. About 10%, calcium chloride is also added to the solution. Here calcium chloride helps to lower the melting point of the mixture.

Temperature : Above 1270 K.

The chemical reactions involved in this process are as follows:

Ionisation of alumina: $Al_2O_3 \rightarrow 2Al^{3+} + 3O^{2-}$

Reaction at cathode: $2Al^{3+}$ (melt) $+ 3e^- \rightarrow Al_{(l)}$

Reaction at anode: $2O^{2-}$ (melt) $\rightarrow O_2 + 3e^-$

- Since carbon acts as anode the following reaction also takes place on it.

$$C_{(s)} + O^{2-} \text{ (melt)} \rightarrow CO + 2e^-$$
$$C_{(s)} + 2O^{2-} \text{ (melt)} \rightarrow CO_2 + 4e^-$$

- Due to the above two reactions, anodes are slowly consumed during the electrolysis.
- The pure aluminium is formed at the cathode and settles at the bottom.
- The net electrolysis reaction can be written as follows:

$$4Al^{3+} \text{ (melt)} + 6O^{2-} \text{ (melt)} + 3C_{(s)} \rightarrow 4A_{(l)} + 3CO_{2(g)}$$

(OR)

(b) (i) Write a note on Fisher Tropsch synthesis. [3Marks]

The reaction of carbon monoxide with hydrogen at a pressure of less than 50 atm using metal catalysts at 500-700 K yields saturated and unsaturated hydrocarbons.

$$nCO + (2n+1)H_2 \longrightarrow C_nH_{(2n+2)} + nH_2O$$
$$nCO + 2nH_2 \longrightarrow C_nH_{2n} + nH_2O$$

(ii) Give the uses of Argon. [2Marks]

- Mixed with 20.06% nitrogen and it is used in gas filled electric lamps.
- It is also used in radio valves and tubes.
- Argon prevents the oxidation of hot filament and prolongs the life in filament bulbs.

35.(a) (i) What is lanthanide contraction? What are the causes of Lanthanide Contraction. [3Marks]

Lanthanoid contraction:

As we move across 4f series, the atomic and ionic radii of lanthanoids show gradual decrease with increse in atomic number. This decrese in ionic size is called ***lanthanoid contraction.***

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Consequences of lanthanoid contraction:

1. Basicity differences:

As we from Ce^{3+} to Lu^{3+} , the basic character of Ln^{3+} ions decrease. Due to the decrease in the size of Ln^{3+} ions, the ionic character of Ln-OH - bond decreases (covalent character increases) which results in the decrease in the basicity.

2. Similarities among lanthanoids:

In the complete f - series only 10 pm decrease in atomic radii and 20 pm decrease in ionic radii is observed. because of this very small change in radii of lanthanoids, their chemical properties are quite similar.

The elements of the second and third transition series resemble each other more closely than the elements of the first and second transition series.

(ii) Write Bragg's Equation? [2Marks]

The fundamental equation that gives a simple relation between the wavelength of the X-rays, the interplanar distance in the crystal and the angle of reflection is known as Bragg's equation.

$$n\lambda = 2d \sin\Theta$$

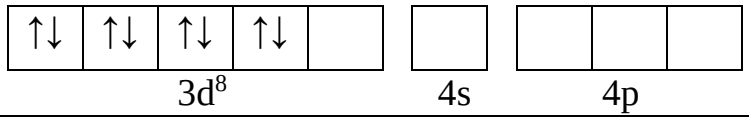
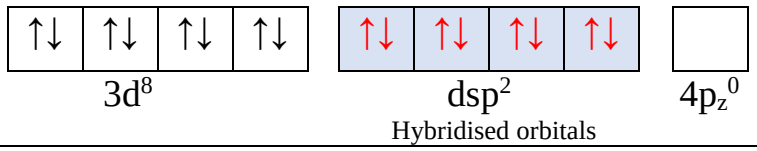
where **n** is the order of reflection, **λ** is the wavelength of X-rays, **d** is the interplanar distance in the crystal, **Θ** is the angle of reflection

(OR)

(b) Based on VB theory explain why [Cr(NH3)6]3+ is Paramagnetic While [Ni(CN)4]2- is diamagnetic. [5Marks]

Complex	**$[Cr(NH_3)_6]^{3+}$**
Central metal ion and its electronic configuration	Cr^{3+} : $3d^3\ 4s^0$
Outer orbitals of metal atom/ion	↑ ↑ ↑ | | | 3d³ | 4s | 4p
Nature of ligand	NH_3 is strong field ligand. However, no pairing of 3d electrons in the metal. **(Refer CFSE)**
Outer orbital of metal atom/ion in presence of ligand	↑ ↑ ↑ | | | 3d³ | 4s | 4p
Hybridisation	Coordination number – 6 Hybridisation – d^2sp^3
Hybridised orbitals of the metal atom in the complex	↑ ↑ ↑ (3d³) ↑↓ ↑↓ ↑↓ ↑↓ ↑↓ ↑↓ (d^2sp^3 Hybridised orbitals)
Geometry	Octahedral
Magnetic property	No of unpaired electron = 3, Hence paramagnetic
Magnetic moment	$\mu_s = \sqrt{n(n+2)} = \sqrt{3(3+2)}$ = **3.872 BM**

Complex	**$[Ni(CN)_4]^{4-}$**
Central metal ion and its electronic configuration	Ni^{2+} : $3d^8\ 4s^0$
Outer orbitals of metal atom/ion	↑↓ ↑↓ ↑↓ ↑ ↑ | | | 3d⁸ | 4s | 4p

Nature of ligand	CN is strong field ligand causes the pairing of 3d electrons in the metal.
Outer orbital of metal atom/ion in presence of ligand	
Hybridisation	Coordination number – 4 Hybridisation – dsp^2
Hybridised orbitals of the metal atom in the complex	
Geometry	Square planar
Magnetic property	No of unpaired electron = 0, Hence diamagnetic
Magnetic moment	$\mu_s = \sqrt{n(n+2)} = \sqrt{0(0+2)} = \mathbf{0}$

36.(a) **Define Half life of a reaction. Show that for a first order reaction half life is independent of initial concentration. [5Marks]**

- The half life of a reaction is defined as the time required for the reactant concentration to reach one half its initial value.
- For a first order reaction, the half life is a constant i.e., it does not depend on the initial concentration. The rate constant for a first order reaction is given by

$$k = \frac{2.303}{t}\log\frac{[A_0]}{[A]}$$

$$\text{at } t = t_{1/2} \; ; \; [A] = \frac{[A_0]}{2}$$

$$k = \frac{2.303}{t_{1/2}}\log\frac{[A_0]}{[A_0]/2}$$

$$k = \frac{2.303}{t_{1/2}}\log 2$$

$$k = \frac{2.303 \times 0.3010}{t_{1/2}} = \frac{0.6932}{t_{1/2}}$$

$$t_{1/2} = \frac{0.6932}{k}$$

(OR)

(b) (i) **Derive Henderson – Hasselbalch Equation. [3Marks]**

- The concentration of hydronium ion in an acidic buffer solution depends on the ratio of the concentration of the weak acid to the concentration of its conjugate base present in the solution i.e.,

$$[H_3O^+] = K_a \frac{[acid]_{eq}}{[base]_{eq}} ;$$

- The weak acid is dissociated only to a small extent. Moreover, due to common ion effect, the dissociation is further suppressed and hence the equilibrium concentration of the acid is nearly equal to the initial concentration of the unionised acid. Similarly, the concentration of the conjugate base is nearly equal to the initial concentration of the added salt.

$$[H_3O^+] = K_a \frac{[acid]}{[salt]}$$

- Here [acid] and [salt] represent the initial concentration of the acid and salt, respectively used to prepare the buffer solution Taking logarithm on both sides of the equation

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$$\log[H_3O^+] = \log K_a + \log\frac{[acid]}{[salt]}$$

reverse the sign on both sides $-\log[H_3O^+] = -\log K_a - \log\frac{[acid]}{[salt]}$

We know that, $pH = -\log[H_3O^+]$ and $pK_a = -\log K_a$

$pH = pK_a - \log\frac{[acid]}{[salt]}$; $\Rightarrow pH = pK_a + \log\frac{[salt]}{[acid]}$

similarly for a basic buffer, $\Rightarrow pOH = pK_b + \log\frac{[salt]}{[base]}$

(ii) Can Fe^{3+} oxidises bromide to bromine under standard conditions. [2Marks]

Given $E^0_{Fe^{3+}/Fe^{2+}} = 0.771\,V$; $E^0_{Br_2/Br^-} = 1.09\,V$

Required half-cell reaction,

$2Br^- \longrightarrow Br_2 + 2e^-$ (E°_{ox}) = -1.09V

$2Fe^{3+} + 2e^- \longrightarrow 2Fe^{2+}$ (E°_{red}) = +0.771V

$2Fe^{3+} + 2Br^- \longrightarrow 2Fe^{2+} + Br_2$ (E°_{cell}) = ?

$(E^\circ_{cell}) = (E^\circ_{ox}) + (E^\circ_{red})$ = -1.09 + 0.771 = -0.319V

E°_{cell} is –ve; ΔG is +ve and the cell reaction is non-spontaneous. Hence Fe^{3+} cannot oxidises Br^- to Br_2.

37. (a) Differentiate Physisorption and Chemisorption. [5Marks]

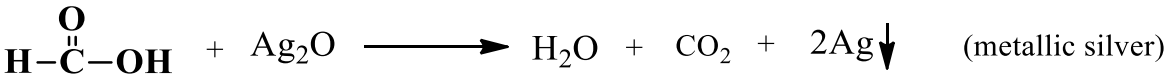
S. No.	Chemical adsorption or Chemisorption or Activated adsorption	Physical adsorption or van der waals adsorption or Physisorption
1	It is very slow	It is instantaneous
2	It is very specific depends on nature of adsorbent and adsorbate.	It is non-specific
3	Chemical adsorption is fast with increase pressure, it can not alter the amount.	In Physisorption, when pressure increases the extent of adsorption increases.
4	When temperature is raised chemisorption first increases and then decreases.	Physisorption decreases with increase in temperature.
5	Chemisorption involves transfer of electrons between the adsorbent and adsorbate.	No transfer of electrons
6	Heat of adsorption is high i.e., from 40-400kJ/mole.	Heat of adsorption is low in the order of 40kJ/mole.
7	Monolayer of the adsorbate is formed.	Multilayer of the adsorbate is formed on the adsorbent.
8	Adsorption occurs at fixed sites called active centres. It depends on surface area	It occurs on all sides.
9	Chemisorption involves the formation of activated complex with appreciable activation energy.	Activation energy is insignificant.

(OR)

(b) (i) Explain the reducing Property of formic acid? [3Marks]

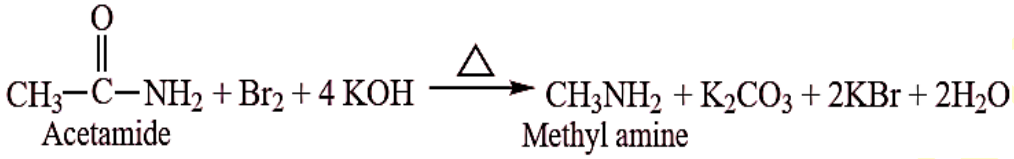
- Formic acid (HCOOH) is unique because it contains both an **aldehyde group** and **carbonyl group**.

- Hence it can act as **reducing agent**.
- Formic acid reduces ammoniacal silver nitrate solution (**Tollen's reagent**) to metallic silver.



(ii) Write a note on Haffmann's degradation reaction? [2Marks]

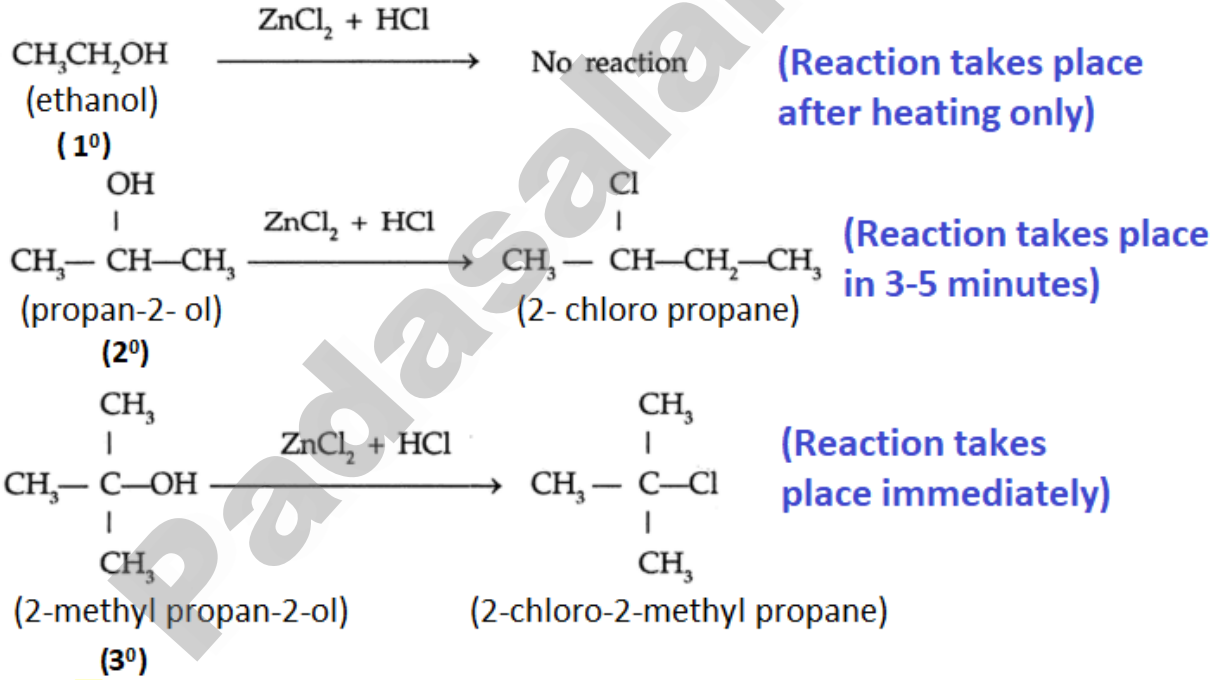
Amides reacts with bromine in the presence of caustic alkali to form a **primary amine** carrying one carbon less than the parent amide.



38.(a) Explain Lucas test. [5Marks]

When alcohols are treated with Lucas agent (a mixture of concentrated HCl and anhydrous $ZnCl_2$) at room temperature.

- Tertiary alcohols react immediately to form a turbidity due to the formation of alkyl chloride which is insoluble in the medium.
- Secondary alcohols react within 10 minutes to form a turbidity of alkyl chloride.
- Where primary alcohols do not react at room temperature.



(OR)

(b) (i) Give any three difference between DNA and RNA. [3Marks]

DNA	RNA
It is mainly present in nucleus, mitochondria and chloroplast	It is mainly present in cytoplasm, nucleolus and ribosomes
It contains deoxyribose sugar	It contains ribose sugar
Base pair A = T. G ≡ C	Base pair A = U. C ≡ G
Double stranded molecules	Single stranded molecules
It's life time is high	It is Short lived

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DNA	RNA
It is stable and not hydrolysed easily by alkalis	It is unstable and hydrolyzed easily by alkalis
It can replicate itself	It cannot replicate itself. It is formed from DNA.

(ii) What are Antibiotics? [2Marks]

The medicines that have the ability to kill the pathogenic bacteria are grouped as ***antibiotics***.

Antibiotic drugs: amoxicillin, ampicillin, cefixime, cefpodoxime, erythromycin, tetracycline etc..

PREPARED BY,

Mr. S.JOHNSON., M.Sc., M.Sc., B.Ed.,

PGT – CHEMISTRY

SACRED HEART MAT. HR. SEC. SCHOOL,

SHOLINGANALLUR, CHENNAI – 600119.



Mr. S.JOHNSON, M.Sc., M.Sc., B.Ed., Padasalai.Net